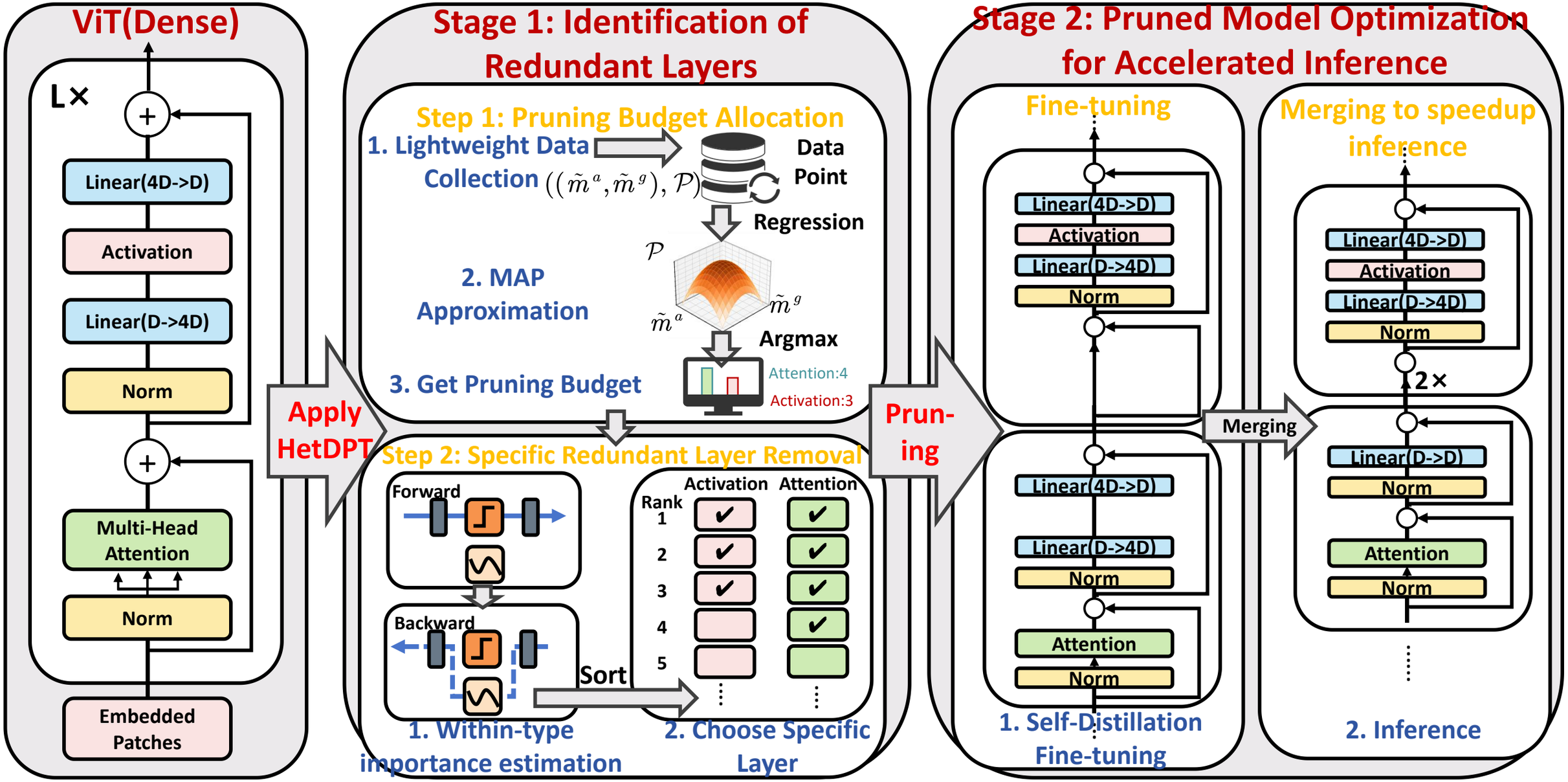
ViT(Dense)
L×
Linear(4D->D)
Activation
Linear(D->4D)
Norm
Multi-Head Attention
Norm
Embedded Patches
Apply HetDPT
Stage 1: Identification of Redundant Layers
Step 1: Pruning Budget Allocation
1. Lightweight Data Collection
Data Point
Regression
2. MAP Approximation
Argmax
3. Get Pruning Budget
Attention:4
Activation:3
Step 2: Specific Redundant Layer Removal
Forward
Backward
Activation
Attention
Rank
Sort
1. Within-type importance estimation
2. Choose Specific Layer
Prun-ing
Stage 2: Pruned Model Optimization for Accelerated Inference
Fine-tuning
Merging to speedup inference
Merging
2×
1. Self-Distillation Fine-tuning
2. Inference